

Witness-Guided Repair and Dual-Gate Checks for LLM-Generated Guard Specifications

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Abstract—Large language models synthesise code that often looks correct under everyday unit tests yet still mishandles thin specification regions—especially catch-all residuals and ordered overlaps. Consider a tier classifier whose OTHERWISE branch must return Review: an LLM candidate can pass every Reject/Pass sample while silently promoting the mid-band to Pass. A release dashboard that tracks mean pass rate alone will ship that defect.

We study a bounded acceptance check: release only when every specification-derived witness matches *and* a structural screen clears ($\text{Accept} \Leftrightarrow \text{Conf}=1 \wedge \text{Screen}=\text{pass}$). HSP-Agile implements this dual-gate check on SOFL/FSF with first-match SMT witnesses, structured scenario-aware repair feedback, and a configurable static safety screen. Here FAR is the fraction of faulty mutants that a mode incorrectly releases; PDR is the complementary rejection rate.

When repair still has headroom, scenario-aware feedback improves mean conformance over test-only feedback by +7.7 pp (95% CI [2.3, 13.6]); all 14 wins repair catch-all residuals. On the constructed first-order mutant population, the dual-gate check reduces FAR from 8.8% to 5.0% (PDR 91.2% to 95.0%). At matched budget $K=3$, equal- K Conf. places M_{eq} at 100.0% vs. B2 97.5% (+2.5 pp; 3/0/117). Gemini hard-seed runs support the feedback effect, while GPT-4o/Claude and public-corpus results reveal frequent saturation. Test-feedback is therefore the default; the additional gate is justified when its bounded witness and mutant criteria match the release context.

Index Terms—acceptance checking, mutant rejection, specification-guided repair, ordered guards, LLM synthesis, SOFL, SMT, false accept rate

I. INTRODUCTION

A release dashboard can look healthy while a residual specification region is already broken. LLMs synthesise code that passes busy unit tests yet mishandles catch-all OTHERWISE bands and ordered overlaps that everyday suites never sample [1]–[3]. SOFL/FSF makes those regions explicit as first-match scenarios [4]–[6]. We ask whether a team can *refuse* to ship under an acceptance-safety bar that uses the specification itself as evidence—without full deductive verification.

Acceptance criterion. Release requires every specification-derived witness to match *and* a structural screen to clear: $\text{Accept} \Leftrightarrow \text{Conf}=1 \wedge \text{Screen}=\text{pass}$. We refer to this directly as a **dual-gate acceptance check**. It is a bounded operational criterion over the generated witnesses and configured screen, not a proof of program safety. HSP-Agile realises it on SOFL/FSF with SpecIR witnesses, structured scenario-aware feedback under headroom, and a configurable static screen. FAR (False Accept Rate) is the fraction of faulty mutants that a

mode incorrectly releases; PDR is the complementary rejection rate.

Primary vignette (catch-all). A tier classifier rejects negative scores and passes scores at or above a floor; everything else falls into OTHERWISE and must return Review. An LLM candidate collapses that residual to Pass. Busy Reject/Pass tests succeed; the mid-band never reaches a human. Typed IR names the residual scenario (others)—the same regime as all 14 E6 Conf. wins over test-only feedback (0/14 ordering repairs). The intro uses a three-scenario skeleton with residual S_3 ; BENCH-120 hard tasks scale the same first-match shape to eight scenarios, so coded wins refer to residual index S_8 .

Secondary vignette (ordering). Ordered tax/sensor overlaps motivate first-match witnesses Φ_i and prevention mutants that reorder or drop preemption; they are not the E6 win set. A public API ACL reconstruction (`enforce_api_acl`) fails on OTHERWISE while guest/admin tests pass—Accept stays red.

Contributions.

- A first-match witness construction that exposes residual and precedence regions missed by ordinary test suites.
- A controlled comparison showing that scenario-aware repair feedback improves mean conformance by +7.7 pp over test-only feedback at matched $K=3$; all 14 wins repair catch-all residuals.
- A dual-gate prevention evaluation in which FAR falls from 8.8% to 5.0%, together with cross-model and public-corpus evidence that identifies when the additional checking is worth its cost.

The evidence therefore separates mechanism (E6), prevention (E2), and deployment under model or corpus saturation.

II. RELATED WORK

Closest LLM+verifier systems differ on whether witnesses respect first-match order, whether repair carries scenario structure, and whether prevention (PDR/FAR) is measured (Table I).

Proof-oriented oracles. DafnyBench and Clover target proof discharge [7], [8]; success is a theorem, not first-match region coverage. Our Conf. metric is witness agreement under SpecIR, not a proof certificate. CEGIS-style SMT loops are

TABLE I
POSITIONING (TICK = DESIGN GOAL).

Approach	Ordered Φ_i	Typed IR	Prev.	Gate
Clover / DafnyBench	—	partial	—	proof
Self-Debug / Reflexion	—	—	—	—
ChatRepair / APR	—	—	—	tests
B2 test-feedback	—	—	—	tests
B6 VerifierLoop-FSF	✓	—	—	cex
HSP-Agile / M	✓	✓	✓	Accept

closer in spirit, but typically synthesise programs or invariants from logical specs rather than ordered-guard FSF with typed repair records.

Verbal and agentic repair. Self-Refine, Self-Debug, and Reflexion raise generation success via critique or verbal memory [9]–[11]. That channel is LLM-as-judge style: useful, but not bound to an executable first-match oracle—the judge may omit the catch-all scenario index. CodeT and coding agents optimise repository tasks without an FSF oracle [12]–[16]. LLM-based APR patches failing tests [17]–[21]; we evaluate unstructured test feedback as B2 and isolate typed IR in E6.

Adjacent notations and SOFL. Alloy and TLA+ encode structured constraints [22], [23]; SpecIR adapters are the intended bridge when guards are first-match and SMT-encodable. Contracts and proof-oriented LLM loops occupy complementary points [24], [25]. SOFL has a long industrial and teaching lineage [4]–[6]; railway and crossing cases motivate ordered-guard contracts but do not release public .asf1 dumps [26], [27]. Related SOFL fault-prevention work studies specification-to-program defects [28]; we target LLM repair under an executable SpecIR oracle. HSP-Agile combines first-match SMT witnesses, scenario-aware repair feedback, and a dual-gate release check evaluated with PDR/FAR.

III. APPROACH

A. SpecIR and first-match witnesses

For task t with ordered scenarios $\langle s_1, \dots, s_n \rangle$, SpecIR exposes first-match regions as a shared hinge across notations (SOFL/FSF, Mini-Z, Mini-StateMachine adapters). Each scenario $s_i = (G_i, P_i)$ pairs a guard with a postcondition; FSF selects the least i with $G_i(\mathbf{x})$ and requires $P_i(\mathbf{x}, \mathbf{y})$. For scenario s_i , the witness generator solves

$$\Phi_i \triangleq \hat{g}_i(\mathbf{x}) \wedge \bigwedge_{j < i} \neg \hat{g}_j(\mathbf{x}) \quad (1)$$

with Z3 [29], [30], yielding suite $\mathcal{D}(t)$. Without the negated higher guards, a model of g_i alone may still belong to s_j for $j < i$. An *others* witness uses $\bigwedge_i \neg \hat{g}_i$ when nonempty; a prior *others*-witness bug (*others* encoded without $\neg$ prior guards) inflated false failures and is disclosed as a measurement correction—we cite fixed-oracle runs as primary. Soundness is conditional on a faithful SMT encoding of SpecIR guards (integer/boolean fragments in our corpora). Ordinary random suites can undersample thin Φ_i regions, allowing residual or ordering bugs to survive green CI [3], [31].

B. Catch-all repair step

Return to the tier classifier: on a mid-band input the buggy candidate returns Pass where the oracle requires Review. The structured scenario-aware feedback record (implemented as a Semantic Feedback IR) contains:

```
{"violation_type": "others", "scenario_index": 2,
 "constraint_text": "OTHERWISE residual",
 "expected": "Review", "observed": "Pass",
 "suggested_fix": "return Review on catch-all"}
```

FeedbackRenderer projects this JSON into the repair prompt. E6 isolates `test_only` / `test_expected` / `semantic_ir` surfaces under identical K . (A secondary ordering example uses inputs such as $(-3, -10)$ with `violation_type:ordering`; that trail motivates Φ_i and E2 mutants, not the E6 win profile.)

C. Scenario-aware feedback and acceptance

The record carries: `violation_type` (ordering, postcondition, *others*), `scenario_index`, `constraint_text`, concrete inputs, `expected/observed` outputs, and an optional `suggested_fix` hint. Conformance and release:

$$\text{Conf}(c, t) = \frac{1}{|\mathcal{D}(t)|} \sum_{\mathbf{x} \in \mathcal{D}(t)} \mathbf{1}[f_c(\mathbf{x}) = g_t(\mathbf{x})], \quad (2)$$

$$\text{Accept}(c, t) \iff \text{Conf}(c, t) = 1 \wedge P^H(c) = 0. \quad (3)$$

Mode B2 uses unstructured test pass/fail only—no scenario index, no constraint text. Controlled E6 isolates feedback content under full M at matched $K=3$: variant A is test-only; B adds expected outputs; C is the full Semantic IR. Equal- K mode $M_{\text{eq}}(K=3, \text{semantic_ir}, \text{advisory gate})$ is the matched-budget Conf. ranking vs. B2. Strengthened E1 ($K=5$, `execution_trace_matched`, advisory gate) is a multi-factor bundle used as a near-ceiling ranking check. E14 further shows that length-matched execution traces can outperform `semantic_ir` on mean Conf. (85.4% vs. 75.1%); the E6 claim is therefore relative to unstructured test feedback.

D. Pipeline

Algorithm 1 summarises one task. Synthesise from FSF text $\rightarrow$ check on $\mathcal{D}(t) \rightarrow$ if not Accept, render Feedback IR and repair, up to budget K ; on exhaustion return $\arg \max_k \text{Conf}(c_k, t)$. Prevention evaluation injects specification-confusion and implementation-screening mutants and scores PDR/FAR on the faulty set. The second gate is instantiated by 14 AST-level safety smells derived from SOFL fault-prevention patterns, including inverted priority, missing preemption, and unconditional success. This catalogue is a reproducible screen instance rather than a completeness claim: any severity-tagged static analyser can replace it without changing the witness check. We therefore attribute conformance changes to witnesses and repair. E2 observes the screen only within the bundled dual gate, while the reported ablation measures conformance rather than a screen-only prevention effect. Overlap density is an engineering hinge for witness generation: E3/E10 tertiles are non-monotone, so overlap alone is not a deployment trigger. The deployable levers are feedback typing (E6) and the Accept/prevention gate (E2).

Algorithm 1 Witness-guided repair with dual-gate acceptance

Require: FSF task t , budget K , feedback mode m

- 1: $\mathcal{D}(t) \leftarrow \text{SpecIR witnesses via Z3}$
- 2: $c^* \leftarrow \perp$; $best \leftarrow 0$
- 3: **for** $k = 1$ to K **do**
- 4: $c \leftarrow \text{synthesise / repair}(t, \text{feedback}_{k-1})$
- 5: $conf \leftarrow \text{Conf}(c, t)$ on $\mathcal{D}(t)$
- 6: **if** $conf > best$ **then** $c^* \leftarrow c$; $best \leftarrow conf$
- 7: **end if**
- 8: **if** $\text{Accept}(c, t)$ **then return** c
- 9: **end if**
- 10: $\text{feedback}_k \leftarrow \text{render}(m, \text{failures on } \mathcal{D}(t))$
- 11: **end for**
- 12: **return** c^* ▷ best-effort argmax Conf.

IV. EXPERIMENTAL SETUP

Benchmarks. **BENCH-120:** synthetic hard FSF (5 inputs, 8 scenarios) retained by a Z3 overlap filter; primary corpus for E1/E6/E12. **E10:** removes the filter ($n=100$) to bound filter bias. **Industrial-pattern** ($n=31$): practitioner-style decision tables. **Desensitized published-industrial** ($n=28$): FSF reconstructed from published railway/ATM/banking SOFL cases [5], [26], [27]. **GitHub harvest** ($n=48$) and **HKCA09→FSF** ($n=35$) probe transfer to public specifications. Vendor .asf1 dumps are not redistributed; an import path exists for future NDA pilots (Table II).

Models, modes, and metrics. B1 one-shot; B2 test-feedback ($K=3$); M (strengthened bundle, $K=5$); M_eq (equal- K , $K=3$, semantic_ir, advisory). E6 variants A/B/C hold full M fixed and vary only feedback surface. Metrics: mean Conf.; Strict / Accept; PDR/FAR on mutants; paired wins/losses/ties with bootstrap CIs where claimed. The controlled E6 mechanism test uses ecnu-plus (Qwen-family). Existing replications include gemini-2.5-flash on hard repair seeds and GPT-4o, Claude, and DeepSeek on industrial-pattern tasks; these runs test transfer and saturation rather than replace the controlled E6 contrast. Repair temperature $T_{\text{repair}}=0$; witness budget up to 32 formal cases per task in strengthened runs.

Mutants (E2). The operators model recurring guard-generation errors: swapping or negating a scenario, reversing an output, dropping a condition, shifting a boundary, flipping a branch condition, reordering returns, dropping a branch, or reversing a dependency. Each operator makes one first-order change to a known-correct specification or implementation; uncompileable mutants are discarded. These faults exercise first-match semantics and postconditions, not just the 14 screen patterns. The implementation track contains $n=852$ mutants; pooling it with specification-confusion gives $n=1704$. PDR is the fraction of faulty candidates rejected; FAR is the fraction of faulty mutants incorrectly released under the release predicate.

Protocol. E6 isolates feedback content; E2 evaluates prevention; M_eq gives the equal- K ranking; strengthened E1 supplies near-ceiling context.

TABLE II
EVALUATION CORPORA AND THEIR ROLES.

Corpus	n	Role
BENCH-120	120	Primary E1/E6/E12 (overlap-filtered)
E10 unfiltered	100	Filter-bias bound
Industrial-pattern	31	Practitioner-style decision tables
Desens. published	28	Reconstructed SOFL; all-mode 100%
GitHub harvest	48	Public .asf1; B2=M_eq
HKCA09→FSF	35	Public SOFL modules; B2=M_eq

TABLE III
FIELD STRUCTURE UNDER HARD COMBO SEEDS (GEMINI; $n=120$ CELLS).

FULL - X	Δ (pp)	W/L/T	Excl. 0
test_only	+33.2	47/8/65	Yes
test_expected	+32.6	47/8/65	Yes
ir_nl_only	+28.0	46/12/62	Yes
ir_no_scenario_id	-0.9	22/17/81	No
ir_no_expected	+5.0	29/17/74	No

V. RESULTS

A. RQ1: Scenario-aware repair feedback (E6)

Figure 1 and Table IV hold K and tasks fixed under full M ($K=3$, $n=120$). Variant C (full Semantic IR) reaches 86.9% mean Conf.—**+7.7 pp** over test-only (A, 79.1%) and +8.0 pp over test+expected (B, 78.9%). Paired: **14/4/102**; 95% CI [2.3, 13.6] pp; Wilcoxon $p=0.018$. Most tasks tie near ceiling; the 14 wins are a rescue regime: 13/14 have test-only Conf=0, and IR records name catch-all scenario others with arithmetic/output mismatches (Table V). The four losses are mixed output/arithmetic residuals. Ordering/preemption defects still motivate Φ_i witnesses and E2 mutants. E14: semantic_ir 75.1% vs. execution_trace_matched 85.4% (5/19/96)—E6 is relative to unstructured feedback. Hard combo seeds on gemini-2.5-flash (pooled $n=80$): FULL vs. test-only +26.5 pp (CI excl. 0). Field ablation (Table III): FULL beats NL-only dumps of the same facts (+28.0 pp; CI excl. 0), while dropping any single IR field yields CIs that include 0—structure vs. unstructured text, not one-field necessity.

B. RQ2: Prevention under the dual gate (E2)

On implementation-screening mutants ($n=852$), M reaches PDR 95.0% / FAR 5.0% vs. B2 91.2% / 8.8%; paired bootstrap 95% CIs on the gaps exclude zero (Fig. 2). The absolute PDR lift is +3.8 pp on this mutant population. Fixed-oracle Conf. ablations vs. strengthened M: removing repair -25.8 pp; removing formal checker -17.5 pp; removing pattern guard 0.0 pp on Conf. under an advisory gate. These ablations isolate conformance effects, whereas E2 evaluates the bundled dual gate; they do not separately identify the screen’s causal contribution to prevention.

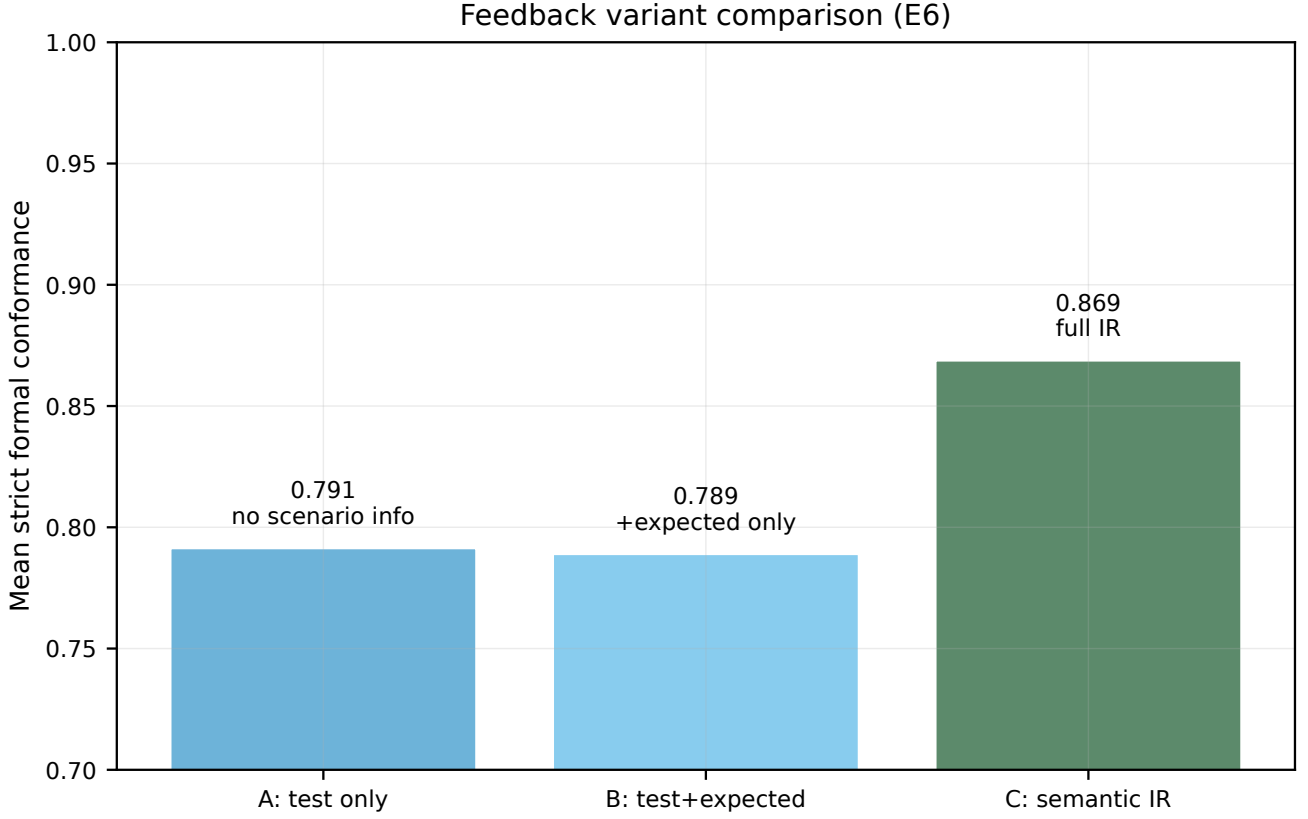


Fig. 1. E6 feedback variants under full M ($K=3$, $n=120$). Variant C (full Semantic IR) +7.7 pp vs. A; CI excludes 0.

TABLE IV
E6 PAIRED STATS, WIN PROFILE, AND EQUAL- K CONF. RANKING.

Contrast / mode	W/L/T or n	Conf. / Δ	95% CI
E6 C-A	14/4/102	+7.7 pp	[2.3, 13.6]
E6 C-B	14/4/102	+8.0 pp	[2.2, 14.0]
Win profile	13/14 zero-A	others/arith.	—
Equal- K B2	120	97.5%	—
Equal- K M_eq	120	100.0% (+2.5)	3/0/117

TABLE V
DEFECT-CLASS $\leftrightarrow$ EVIDENCE MAP.

Defect class	Primary evidence	Supports
Order / preemption	Φ_i ; E2 mutants	dual-gate prevention
others/arith. rescue	E6 win coding	feedback mechanism
Saturated easy tasks	pilots ($n \geq 28$)	default B2

TABLE VI
CROSS-MODEL INDUSTRIAL-PATTERN RESULTS ($n=31$; MEAN CONF. %).

Model	B1	B2	M
GPT-4o	95.5	100.0	95.5
Claude 4.6	95.5	100.0	95.5
DeepSeek	52.7	80.7	72.5

TABLE VII
E1 NEAR-CEILING RANKING (FIXED ORACLE; $K=5$ BUNDLE FOR M).

Mode	Mean Conf.	Strict	Mean LLM calls
B1	84.2%	84.2%	1.00
B2	98.3%	98.3%	~ 1.18
M ($K=5$ bundle)	100.0%	100.0%	~ 1.16

C. RQ3: Cross-model behavior and matched-budget evidence

The catch-all failure is not specific to one model family, but the benefit of additional feedback depends on remaining repair headroom. On `gemini-2.5-flash`, frozen hard seeds yield +26.5 pp for scenario-aware feedback over test-only feedback (pooled $n=80$, CI excludes zero). On industrial-pattern tasks, GPT-4o and Claude both reach 100.0% with B2 while the fuller mode reaches 95.5%; DeepSeek retains more headroom (80.7% for B2 and 72.5% for M). These results show that

stronger models can saturate under ordinary test feedback; they support conditional deployment, not a universal gain from the full pipeline.

Equal- K Conf. ranking (Table IV): M_eq 100.0% vs. B2 97.5% (+2.5 pp; 3/0/117)—matched budget, not a substitute for E6. Unified external equal- K : GitHub B2=M_eq 89.9%; HKCA09 both 100%; desensitized both 100%. Strengthened E1 ($K=5$ bundle, Table VII) sits near ceiling. E12: B2 reaches 100% on every seed; M mean 99.7% across seeds.

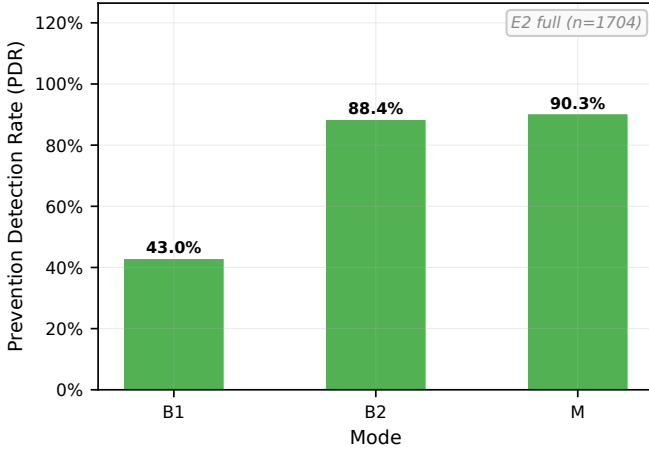


Fig. 2. E2 aggregate PDR by mode.

TABLE VIII
DEPLOYMENT BOUNDARY: RELEASE BAR AND REPAIR HEADROOM.

Signal	Prefer B2	Prefer M
Default	Mean Conf./latency	—
Release	Mean Conf. OK	Accept=1
Prevention	—	FAR $\leq$ 5%
Headroom	Saturated endpoint	Collapsed test-only Conf. (E6)
Overlap tertile	Non-monotone	Not a trigger

TABLE IX
EXTERNAL CORPORA (MEAN CONF. %).

Corpus	B1	B2	M / M_eq
Industrial-pattern ($n=31$, gpt-4o)	95.5	100.0	95.5
Desens. published ($n=28$)	100.0	100.0	100.0
GitHub harvest ($n=48$)	85.8	89.9	89.9
HKCA09 FSF ($n=35$)	74.3	100.0	100.0
Equal-K BENCH-120	—	97.5	100.0

D. RQ4: Deployment boundary and public-corpus transfer

Canonical E3/E10 tertiles are non-monotone (Fig. 3): escalate on Accept/FAR and headroom, not on overlap density alone (Table VIII). On fixed-oracle E1, mean LLM calls for B2 (~ 1.18) and strengthened M (~ 1.16) are comparable; wall-clock means are 9.6 s (B2) vs. 10.5 s (M).

1) *Public and published corpora*: Desensitized published pilot ($n=28$): B1/B2/M_eq all 100.0%. GitHub harvest ($n=48$): B2/M_eq 89.9% vs. B1 85.8%. HKCA09 $\rightarrow$ FSF ($n=35$): B1 74.3% / B2=M_eq 100%. These ranking ties support B2 as the default for mean Conf. Where one-shot saturates, HKCA09 hard-seed repair (wrong_relop: semantic_ir 82.0% vs. test_only 69.3%, +12.7 pp; CI may include 0) is scoped external support. Provenance: reconstructions / public harvest—not proprietary dumps.

2) *Public ACL interception case*: The aggregate results are complemented by `enforce_api_acl`, an ordered API-access contract reconstructed from the public RealSpec/harvest family. The contract denies guests, requires step-up MFA for high

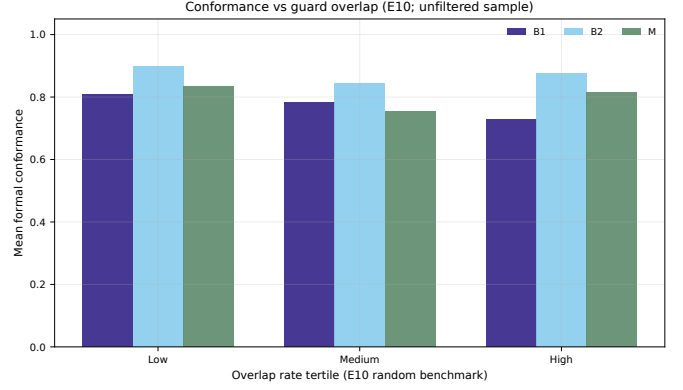


Fig. 3. E10 Conf. vs. overlap tertile (non-monotone).

scope, permits narrow admin/reader bands, and fails closed on OTHERWISE. An LLM-shaped candidate passes the busy guest-deny and admin-permit tests but contains two residual defects: high scope without MFA is permitted, and the catch-all branch fails open. A first-match witness ($role=2, scope=2, mfa=1$) forces the residual branch; the candidate returns permit, so Conf <1 and the dual-gate check refuses release. This reproducible case illustrates the operational target of the method, while remaining smaller than a repository-scale industrial study.

3) *Scope of transfer*: The controlled causal estimate remains E6 on the primary endpoint; Gemini supports the repair mechanism on frozen hard seeds, while GPT-4o and Claude demonstrate saturation on industrial-pattern tasks. Public and reconstructed corpora support the deployment boundary, but they do not constitute repository-scale validation.

VI. DISCUSSION

Practice. Match feedback depth to the release bar: B2 for mean Conf. / latency; M when dual-gate Accept or FAR $\leq$ 5% is required and repair headroom remains (E6 rescue when test-only Conf. collapses). External pilots reinforce default B2 on mean Conf. A practical playbook: (i) B2 inner loop; (ii) pre-release M with Accept; (iii) track FAR on a frozen mutant pack; (iv) escalate if FAR drifts or Accept failures cluster.

Limitations. BENCH-120 is synthetic and overlap-filtered; E10 and public pilots probe, but do not bound, transfer. Finite witnesses approximate Conf.; the others-witness measurement correction previously inflated false failures. LLM non-determinism is mitigated by $T_{\text{repair}}=0$ and paired tests; the $K=5$ E1 bundle combines several mechanisms and is used only as near-ceiling context. Published reconstructions and harvests are not production dumps; vendor NDA .asf1 pilots remain future work. The screen catalogue and mutation operators are author-selected from SOFL fault patterns; E2 therefore estimates rejection on that constructed first-order distribution, not completeness or repository-wide safety. Ceiling effects

on stronger models shrink E6 deltas—report headroom when transferring claims. E14 shows that length-matched execution traces can outperform the tested scenario-aware record on mean Conf.; E6 therefore establishes improvement over unstructured test feedback rather than superiority over every structured feedback design. B2 leads every E12 seed (100%); M mean is 99.7% across seeds.

Reproducibility. Authoritative numbers and run ids ship with the artefact package (AUTHORITATIVE_NUMBERS.md, benchmark JSON, summarisation scripts). Primary claims cite those run directories rather than ad-hoc aggregates.

VII. CONCLUSION

HSP-Agile pairs first-match witnesses, scenario-aware repair feedback, and a dual-gate acceptance check for ordered-guard specifications. The central result is a bounded acceptance check: release only under $\text{Accept} \Leftrightarrow \text{Conf}=1 \wedge \text{Screen}=\text{pass}$. Under headroom, E6 yields +7.7 pp vs. test-only at $K=3$ (CI excludes 0; 14/14 catch-all rescues). E2 supports the gate on the evaluated mutant population at FAR 5.0% vs. 8.8%. Deploy B2 by default; escalate for Accept or FAR $\leq 5\%$ with headroom. Future work includes NDA vendor .asf1 pilots, adaptive K , and SpecIR adapters beyond SOFL/FSF.

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